Johnson&Johnson
MedTech

Butterfly Data & Analytics Strategy

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Executive Summary

Companies that succeed in today's competitive MedTech landscape, are those that have access to large volumes of high-quality data and can turn that data into actionable insights. Project Butterfly represents our first major step toward building a global, AI-driven data ecosystem that will be a competitive advantage for J&J MedTech.

The challenge: we are in the "dark ages". Our fragmented data landscape is holding us back. Siloed systems prevent collaboration, frustrate customers, and limit our ability to scale AI and analytics initiatives. Poor data quality leads to costly mistakes and missed opportunities. We're reacting to problems instead of preventing them.

The solution: The Butterfly Data & Analytics Strategy establishes a unified approach to building and maintaining a strong data foundation across regions and data domains. We will create high-quality data products organized into three categories:

- foundational products: the infrastructure layer including account, contact, product etc.
- analytical products: datasets optimized for exploration and analysis.
- decision products: tools that directly support specific business actions and outcomes.

Our approach: we start with building the data foundation. For this purpose, we build global data models, which represent a common language across the globe. By implementing them in Butterfly, we avoid data siloes and ensure connectivity. In the next step, we will tackle analytics and decision products.

Enabling success: Building the infrastructure is only half the battle. We're establishing a comprehensive but light-weight governance mechanism to ensure our data is meaningful and of high quality. We will also drive a comprehensive change management and adoption program to ensure our organization understands what data is available and uses it effectively.

This is a living document. We will complete the strategy with additional topics as we progress.

Preamble

In a highly competitive MedTech market, winning requires precision and foresight – two qualities that are not possible without data. The companies that succeed are those that have access to large volumes of high-quality data and can turn that data into actionable insights.

Data enables us to:

1. **Know our customers better than anyone else** — anticipating needs and tailoring our offerings.
2. **Optimize performance end-to-end** — creating visibility from R&D to Supply Chain to Commercial teams to drive efficiency and margin improvement.
3. **Unlock new revenue streams** — powering software, robotics, and services that create differentiated value for customers.
4. **Deliver superior outcomes** — enabling truly personalized patient care.

To achieve all of this, we must take care of our data: this means capturing it reliably, structuring it effectively, making it accessible, and continually improving its quality. This is how we build and sustain competitive advantage. Competitors can replicate products, but they cannot easily match the depth, integration, and intelligence of a well-designed data ecosystem.

Project Butterfly is our first step in the journey to build this ecosystem on a global scale.

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Butterfly Data & Analytics Strategy

Building a global connected AI-driven customer experience solution requires a strong data foundation that spans multiple regions and data domains. The Butterfly Data & Analytics Strategy defines how we will build and maintain this foundation to enable CXM functionality and insightful analytics that will drive business value.

1. Problems we're solving

Our fragmented data is increasingly impacting our business performance. With project Butterfly we are addressing these problems. Here are just a few examples:

- **Lack of collaboration due to siloed data**: our teams talk to the same HCPs but have no way of knowing this because their datasets are not connected. This is highly frustrating for our customers and colleagues.
- **Failed analytics/AI projects**: we cannot use AI to its full potential because our underlying data foundation is not strong enough. This leads to many pilot projects that do not scale.
- **Mistakes, confusion, inefficiencies due to low quality data**: we suffer from wrong data about our customers, which leads to wrong decisions, costly remediation and high frustration amongst our colleagues.
- **Costly systems, plethora of reports yielding few insights**: isolated approaches to data and analytics have led to island solutions that are costly to maintain while delivering little value.
- **Reacting to issues instead of preventing them**: because of limited data visibility, we react instead of preventing issues from happening. With a strong data foundation, we can predict problems and identify opportunities in advance.

2. Definitions & Principles

Data Domains

The data we process in Butterfly belongs to two types of domains: foundational and experience specific. The **Foundational Data Domains** we require comprise Account, Contact, Product etc. Data from these domains is required by all Experiences. They therefore need to be managed centrally to ensure connectivity. In some of these domains, such as Contact, we may need to include external data.

In addition, there are **Experience Data Domains** that need to be managed in a federated way within each Experience. Here are some examples:

- Sales: activity, opportunity, account plan
- Marketing: lead, segment, profile
- Customer service: case, queue, entitlement
- Service & Repair: asset, work order, resource, booking
- PR&O: annual needs assessment, engagement

Some of these data domains are mastered outside the Butterfly system (e.g. account and product are mastered in the ERP), some within (e.g. opportunity, case). Each foundational domain needs to be owned by one person to ensure coordination.

Data Products

The goal of the Butterfly data strategy is to provide high-quality data products that enable Experience functionality and insightful analytics. A data product is a finished good, made from raw data that delivers value to specific customers.

We differentiate between three types of data products: foundational, analytical and decision-enabling ones.

I. Foundational data products enable functionality but don't directly drive business outcomes. They can be considered part of the infrastructure layer. Typical users of these products are product engineers. This is the current list of foundational data products in scope:
    - Account: data about the accounts we serve
    - Contact: these are the individuals we connect with across Experiences
    - Consent: data product that holds information about marketing and communication consent
    - Product: the products and services we sell
    - Alignment (aka territory): sales territory information
    - Employee: basic data about our employees
    - Transactional data: this will include sales and gross profit but also orders and invoices.

    These data products make up the infrastructure that the Experiences depend on.

II. Analytical data products make raw data analyzable and enable exploration, analysis and insights generation. Customers for these products are business analysts. Examples:
    - PowerBI dataset with CRM metrics
    - Marketing data mart
    - Customer 360 dataset

III. Decision data products directly support specific decisions, actions, or business outcomes. They are used by business users (managers, reps, executives) and have measurable impact on business performance. Here are a few examples of decision data products:
   - Sales pipeline dashboard
   - Sales vs target dashboard
   - Customer churn predictor
   - Lead scoring mechanism
   - AI copilot for sales meeting prep
   - Case assignment logic

Guiding Principles

Within the Butterfly program, we need to ensure that we think "data first". Too often we treat data as an afterthought and are then faced with broken analytics and costly retrofitting. As a rule of thumb, we need to involve data experts in the design of Experiences to ensure we follow the agreed data strategy.

The data teams need to follow the following guiding principles when designing and building their data products.

Universal Principles that apply to all data products:

- **Clear ownership and accountability**: assign a dedicated owner responsible for the product's success, maintenance, and evolution.
- **Documented and discoverable**: provide clear documentation so users can find, understand, and effectively use the product.
- **Measured for value delivery**: track metrics that demonstrate the product's impact and justify continued investment.

Then there are some principals that differ by data product type.

Foundational data products

- **Design for multiple consumers, not single use cases**: build for reusability across teams and purposes. Avoid over-fitting to one specific need.
- **Prioritize quality and reliability**: establish strict data quality standards, leveraging the enterprise DQI framework, automate monitoring, and handle changes carefully since many things depend on this foundation.
- **Plan for evolution**: build incrementally but with a long-term vision. Business needs change, so include versioning, migration paths, and regular reviews.

Analytical data products

- **Optimize for exploration and flexibility**: enable analysts to query, slice, and investigate data freely without rigid constraints or predetermined paths.
- **Balance performance with completeness**: provide sufficient detail for deep analysis while maintaining query performance that supports interactive exploration.
- **Design for analyst self-service**: create intuitive data structures, clear naming conventions, and accessible documentation that reduce dependency on data engineering support.

Decision Data Products

- **Start with the decision, not the data**: identify the specific decision or action users need to take, then work backwards to what data and insights enable it.
- **Design for your user, not yourself**: deeply understand user workflows, pain points, and skill levels. Co-design with actual users through prototypes and iterations.
- **Deliver insights at the point of decision**: provide information when users need to act, in the tools they already use, with timing that matches decision urgency.
- **Iterate based on outcomes**: treat it as a living product with a roadmap. Gather user feedback continuously and evolve based on how well it drives decisions.

Data Ecosystem

Butterfly CXM requires data products to be delivered via the commercial data ecosystem. Global data sources vary widely, so an integration layer is needed to harmonize this information for the Butterfly Global Core. Regions will remain responsible for data quality, and strong governance is necessary to ensure connectivity and quality management throughout.

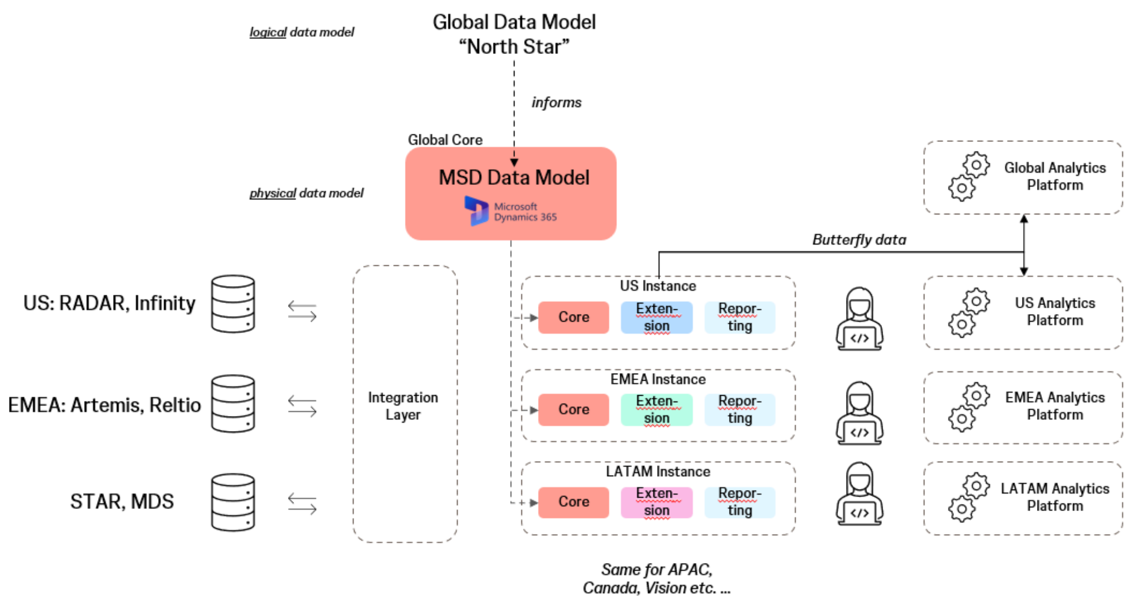
Overview of current MedTech data sources (Vision not yet included):

Data Source	US	EMEA	APAC	Japan	LATAM	Canada
ERP	SAP	JDE, SAP S4 (T1/T2)	SAP MARS	SAP S4 T2, SAP BTB	SAP BTB	SAP BTB
Commercial data lake	Radar / EUSS	Artemis	DataVantage	DataVantage & Denodo	STAR	RADAR/ Specific

						Canada location
Commercial MDM	Reltio (Infinity)	Reltio	MDS	MMS, MDS, SHMMS	[MDS]	N/A
Sales Territory Mgmt	SalesIQ / Zaidyn	SalesIQ	-	-	ZS	SAP BTB – Primary Alignment RADAR Table – Secondary Alignment
Employee	Workday	Workday	Workday	Workday	Workday	Workday

We will also need to ensure data flows back out of Butterfly into regional analytics systems to enable the delivery of analytical and decision data products.

The picture below shows the connections between the logical data model, its physical implementation in MS Dynamics and the required integrations with regional data sources and analytics environments.



3. Building the data foundation

Our first priority is to produce the foundational data products in Butterfly as they enable the rest. Each product needs to go through four phases: define, design, build, govern.

Define

For each foundational data product, we first need to document the business requirement that the product will address. Then we need to define what information it should contain and how its elements should be connected. This is done via a global data model. These data models are the "North Star" that guides the implementation in MS Dynamics.

Design

Once the data model is finalized, we need to implement it in MS Dynamics for the Experiences to access it. The output of this step is a physical manifestation of the data model that can be used in code to build the Experiences.

Designing the native data model needs to be done carefully to ensure that the nuances of the logical data model are captured while the model is optimized for performance in MS Dynamics.

Build & Deploy

Following the design of the foundational data products, we need to build the integration with regional data sources. In some cases, these integrations need to be bi-directional as Butterfly will modify master data, such as contacts.

Once the integrations have been built, data must be loaded into Butterfly. Rigorous data quality checks need to be put in place before any data is released to the Experiences. Foundational data products must be of very high quality.

Govern & Execute

Ongoing governance ensures the continued viability of foundational data products in Butterfly. We will need to monitor data quality (leveraging the enterprise DQI framework) and work on remediation as required. Furthermore, we need to ensure hierarchies and list-of-values (LOV's) are appropriately maintained.

Roles & Responsibilities

Business and Technology teams work together to manage data in Butterfly, each contributing their unique strengths. The Business team focuses on the “what” and the “why”, making sure the data team's efforts generate real business value. The Technology team works on the “how” and ensures that the technical implementation generates this value in the most efficient way possible.

Here are the key roles that we need to assign to talented individuals:

- **Global Business & Technology Data Leads**: own the definition and global execution of this data strategy.
- **Regional Business & Technology Data Leads**: own regional data ecosystems and local business requirements.
- **Global Data Product Managers**: they own global data products and ensure they address business needs.
- **Business Analysts**: they support the creation and maintenance of critical dashboards and reports.
- **Data Enablement Managers**: engage with end-users of all types of data products to ensure they are used well.
- **Data Analysts & Modelers**: design fit-for-purpose data models that bring data products to live.
- **Data Platform Engineers**: responsible for building and managing the data and analytics ecosystem to deliver data products.
- **Data Engineers**: they implement data products in MS Dynamics and build the required integrations with regional data sources.

As the data strategy evolves, this list of roles may change.

Governance and Coordination Forums

Data is a team sport. We need to set up mechanisms to connect these individuals in the most efficient way to govern our data domains and manage the delivery of data products. This is an important element of the strategy that will be refined in future versions.

To address immediate needs, we create two global groups: the **Global Commercial Data Governance Forum** and the **Butterfly Data Product Coordination Forum**.

1. The Global Commercial Data Governance Forum is the group in which global and regional data teams connect to discuss the following topics:
    a. Define and maintain global data models for foundational data domains and ensure cross-domain coordination.

b. Establish data quality standards and guide implementation.
c. Architectural decisions on data platforms and technologies. Data governance would be administered leveraging an enterprise approved data governance tool.
d. Compliance oversight across data domains.

Members: global and regional data leads, data analysts, Experience data leads, architecture lead, compliance lead.

Frequency: monthly (initially bi-weekly).

2. The Experience Data Enablement Forum addresses topics that are relevant to Butterfly Experience teams. This includes the following:
   a. Data model alignment and implementation status.
   b. Manage hierarchies and LOV's in Butterfly.
   c. Manage cross-product data dependencies.
   d. Manage data risks in Butterfly.

   Members: global data leads, Experience data leads, data product owners.

   Frequency: bi-weekly/weekly

As mentioned above, more to come on this topic.

Enable the organization

Building a world-class data foundation is only valuable if our organization knows it exists and uses it effectively. We will establish a comprehensive enablement program led by dedicated Data Enablement Managers who act as the bridge between data products and end users. This includes creating a searchable data product catalog that makes it easy for employees to discover what data is available, understand its quality and limitations, and learn how to access it. We'll develop role-based training programs and self-service resources tailored to different user personas—from sales reps needing quick dashboard insights to analysts building complex reports.

It's crucial that we measure our impact not only by data availability and usage but importantly by the business outcomes it helps drive. We need to show the value of "good data" to business users in their own language.

4. Delivering Analytical & Decision Data Products

We need to be agile in the delivery of analytical and decision data products and should not create bottlenecks that prevent us from getting insights. At the same time, we need to set a framework or standards and governance to ensure they are of high quality.

Each decision data product needs to have a clear purpose and deliver quantifiable business value to a group of stakeholders. We need to set up guardrails to ensure that everything we do with regards to these products delivers business value.

This chapter will be developed further in the coming months.